

SQL fundamentals.

Exercise 1:

Q1. SELECT Statement

Write a SQL query to retrieve all columns from the employee table.

SELECT *

FROM employees-db;

Q2 SELECT DISTINCT

Write a SQL query to find all the unique departments in the employees table:

```
SELECT DISTINCT dept
FROM employees-db;
```

Q3 ORDER BY Statement

Write a SQL query to retrieve all employees 'first and last' names, ORDER BY Salary in DESC order.

```
SELECT first-name,  
       last-name,  
       salary  
FROM employees-db  
ORDER BY salary - DESC;
```

Outcome

first-name	Last-name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Daniel	Clark	53 000

Q4 Limit Statement

Write a SQL query to retrieve the top 5 highest paid employees.

```
SELECT ID,  
        Salary AS highest - Paid  
FROM employees - db  
LIMIT 5 ;
```

Outcome

ID	Salary
6	62 000
3	60 000
7	59 000
1	55 000
9	53 000

Q5 WHERE Statement

Write a SQL query to find employees who work in the IT department

```
SELECT ID,  
       first_name,  
       last_name,  
       department  
FROM employees_db  
WHERE department = IT;
```

Outcome

ID	first_name	last_name	department
1	John	Doe	IT
4	Sarah	Brown	IT
6	Emily	Davis	IT
10	Laura	Hall	IT

Q6 AND Statement

Write a SQL query to find employees who work in the finance department AND have a salary greater than 55,000.

```
SELECT first_name,  
       last_name,  
       ID,  
       department  
       salary  
FROM employees_db  
WHERE department = finance AND salary > 55000;
```

Outcome				
18	first-name	Last-name	department	Salary
9	Robert	Wilson	Finance	59 000
3	Mike	Johnson	Finance	60 000

Q7 OR Statement

Write a SQL query to find employees who work in the HR department OR in the Marketing department.

```
SELECT ID,  
       first-name,  
       last-name,  
       department  
FROM   employees-db  
WHERE  department = HR OR Marketing;
```

Outcome

ID	first-name	last-name	department
2	Jane	Smith	HR
5	David	White	Marketing
8	Jessica	Moore	HR
9	Daniel	Clarke	Marketing

Q6 NOT Statement

Write a SQL query to find employees who do NOT work in the IT department.

```
SELECT ID,  
       first_name,  
       last_name,  
       department  
FROM   employees_db  
WHERE  department != IT;
```

Outcome

ID	first_name	Last_name	department
2	Jane	Smith	HR
3	Mike	Johnson	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
9	Daniel	Clerk	Marketing

Q9. IN statement

Write a SQL query to find employee who are IN the HR, IT, or finance department

```
SELECT ID,  
       first_name,  
       last_name,  
       department  
FROM   employee-db  
WHERE  department IN (HR, IT or finance);
```

Outcome

ID	first-name	Last-name	departement
1	John	Doe	IT
2	Jane	Smith	HR
3	Mike	Johnson	Finance
4	Sarah	Brown	IT
6	Emily	Davis	IT
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
10	Laura	Hall	IT

Q10 Combining Condition

Write a SQL query to find employees who are in the IT department, have a salary greater than 50,000 and are located in New York.

```
SELECT ID,  
       department,  
       Salary,  
       City  
FROM employees - db  
WHERE IT (Salary) > 50000 AND City = New-York
```

Outcome

ID	department	Salary	City
1	IT	55000	NYC
4	IT	58000	NYC